



PELPRENE™ X-157D

TOYOBO MC Corporation - Thermoplastic Copolyester Elastomer

Tuesday, February 25, 2025

General Information			
Product Description			
General Grade			
General			
Generic Name	• Thermoplastic Copolyester Elastomer (TPC-ET)		
Material Status	• Commercial: Discontinued		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Features	• High Hardness		
Uses	• Automotive Applications • Automotive Interior Parts	• General Purpose • Industrial Applications	• Sporting Goods
Forms	• Pellets		
Processing Method	• Injection Molding		
Part Marking Code (ISO 11469)	• >TPC-ET<		

ASTM & ISO Properties ¹			
Physical	Nominal Value	Unit	Test Method
Density / Specific Gravity	1.23	g/cm³	ASTM D792
Density	1.23	g/cm³	ISO 1183
Melt Mass-Flow Rate (MFR) (230°C/2.16 kg)	18	g/10 min	ISO 1133
Molding Shrinkage - Flow ² (2.00 mm)	1.6	%	Internal Method
Molding Shrinkage - Across Flow ² (2.00 mm)	1.7	%	Internal Method
Water Absorption ³ (Equilibrium, 23°C, 50% RH)	0.27	%	ISO 62
Mechanical	Nominal Value	Unit	Test Method
Tensile Modulus	420	MPa	ISO 527
Tensile Stress (Yield)	22.0	MPa	ISO 527-2
Tensile Strength (Break)	38.0	MPa	ISO 37
Tensile Stress (50% Strain)	15.3	MPa	ISO 527
Tensile Elongation (Break)	600	%	ISO 37
Flexural Modulus	290	MPa	ISO 178
Impact	Nominal Value	Unit	Test Method
Charpy Notched Impact Strength	No Break		ISO 179
Hardness	Nominal Value	Unit	Test Method
Shore Hardness			ISO 48-4
Shore A, 15 sec	98		
Shore D, 15 sec	56		

Thermal	Nominal Value	Unit	Test Method
Deflection Temperature Under Load 0.45 MPa, Unannealed	92.0	°C	ISO 75-2
Brittleness Temperature	< -65.0	°C	ISO 974
Melting Temperature	209	°C	Internal Method
CLTE - Flow	1.5E-4	cm/cm/°C	ISO 11359-2
CLTE - Transverse	1.4E-4	cm/cm/°C	ISO 11359-2
Electrical	Nominal Value	Unit	Test Method
Surface Resistivity	3.0E+16	ohms	IEC 62631-3-2
Volume Resistivity	6.7E+10	ohms·m	IEC 62631-3-1

Processing Information			
Injection	Nominal Value	Unit	
Drying Temperature	100 to 120	°C	
Drying Time	> 4.0	hr	
Rear Temperature	200 to 230	°C	
Middle Temperature	220 to 240	°C	
Front Temperature	220 to 240	°C	
Nozzle Temperature	220 to 240	°C	
Mold Temperature	20 to 60	°C	

Notes

- ¹ Typical properties: these are not to be construed as specifications.
- ² 100×100×2mmt, film gate
- ³ 192hr